# INTRODUCTION TO APPLIED NUMERICAL METHODS

## Curve Fitting: Background


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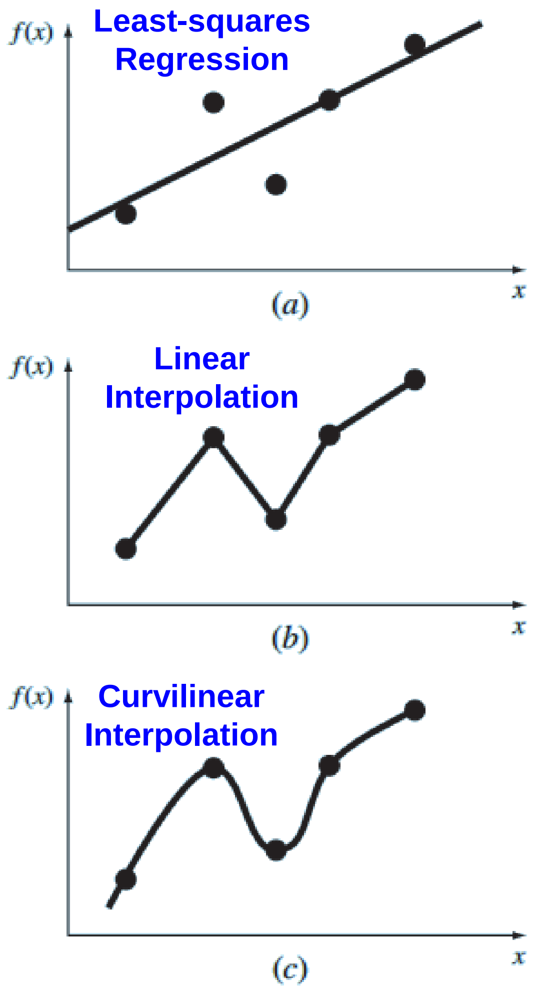
# CURVE FITTING

**Motivation**

- Fit curves to discrete data in order to obtain intermediate estimates (or estimates between the discrete data)

**Curve Fitting Approaches**

- Regression
  - Data exhibit a significant degree of error or "noise" (individual data point incorrect)
  - Derive a single curve that represents the general trend of these data: the pattern of all points
  - Least Squares Regression
- Interpolation
  - Data is very precise
  - Curve pass directly through each point
  - Look-up table (e.g., material properties)
  - Interpolation



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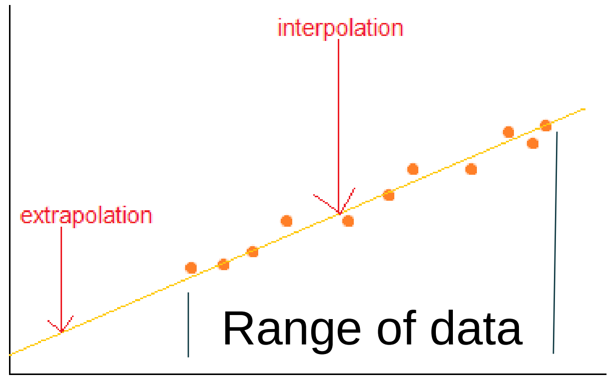
# CURVE FITTING & ENGINEERING PRACTICE

**Applications of Fitting Experimental Data**

- **Trend Analysis**
  - Predict or forecast values of the dependent variables based on the pattern of these data
  - Regression (noisy data) and interpolation (precise data)




- **Hypothesis Testing**
  - Determine the model coefficients of curve fitting that fits the data best
  - Compare the predicted values against the observed values to test the accuracy of the model
  - Compare alternative models (linear vs. nonlinear) and select the best one based on empirical observations

# MATHEMATICAL BACKGROUND

- In course of engineering study, if several measurements are made of a particular quantity, additional insight can be gained by summarizing the data in one or more well chosen statistics that convey as much information as possible about specific characteristics of the data set.

- These descriptive statistics are most often selected to represent
  - The location of the center of the distribution of the data,
  - The degree of spread of the data.

# STATISTICS MEASURES

**Arithmetic Mean ( $\bar{y}$ )**

- The sum of the individual data points ($y_i$) divided by the number of points ($n$)

$$\bar{y} = \frac{\sum y_i}{n}$$


https://www.leanblog.org/2018/01/two-data-points-dont-make-trend-read-beyond-headline/

**Standard Deviation ($s_y$)**

- The measure of the spread of the data

$$s_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}}$$

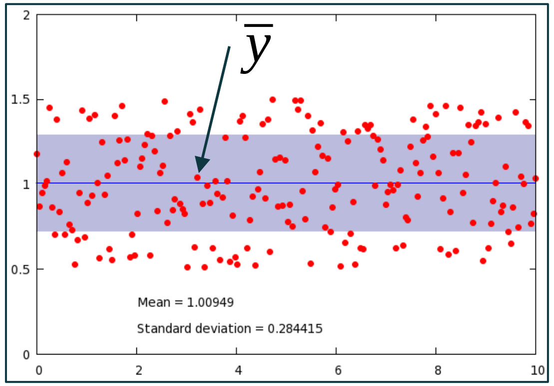
or

$$S_y^2 = \frac{\sum y_i^2 - \left(\sum y_i\right)^2 / n}{n-1}$$

Degrees of freedom

$S_t$: The total sum of the squares of the residuals between the data points and the mean.

$$S_t = \sum (y_i - \bar{y})^2$$




http://www.phyast.pitt.edu/~zov1/gnuplot/html/statistics.html

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# STATISTICS MEASURES

**Standard Deviation ($s_y$)**

- The measure of the spread of the data

$$s_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}}$$

**Variance ($s_y^2$)**

- The measure of the spread of the data
- Square of the standard deviation

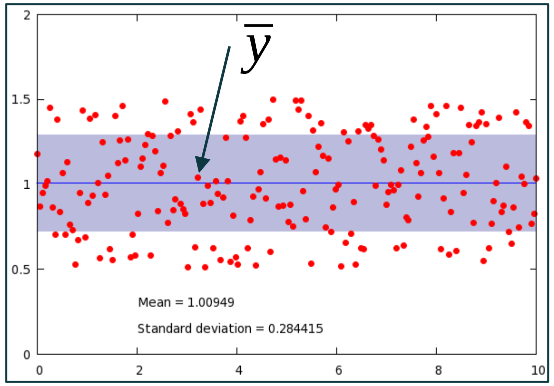
$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{n-1}$$

**Coefficient of Variance (C.V)**

- Ratio = The Standard Deviation : The Mean
- A normalized measure of the spread

$$\text{c.v.} = \frac{s_y}{\bar{y}} 100\%$$


https://www.leanblog.org/2018/01/two-data-points-dont-make-trend-read-beyond-headline/

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# AN EXAMPLE

- 24 Reading of the coefficient of thermal expansion of a structural steel ($\times\ 10^{-6}\ in/(in\ ^{0}F)$)
- Compute the mean, variance, standard deviation, and coefficient of variation for the data in the Table

| | | | | | |
|---|---|---|---|---|---|
| 6.495 | 6.595 | 6.615 | 6.635 | 6.485 | 6.555 |
| 6.665 | 6.505 | 6.435 | 6.625 | 6.715 | 6.655 |
| 6.755 | 6.625 | 6.715 | 6.575 | 6.655 | 6.605 |
| 6.565 | 6.515 | 6.555 | 6.395 | 6.775 | 6.685 |

# AN EXAMPLE

**Compute the mean, variance, standard deviation, and coefficient of variation for the data in the Table**

**Solution**

$$\bar{y} = \frac{\sum y_i}{n} \Rightarrow \bar{y} = \frac{158.4}{24} = 6.6$$

$$s_y = \sqrt{\frac{S_t}{n-1}} = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n-1}}$$

$$s_y = \sqrt{\frac{0.217000}{24 - 1}} = 0.097133$$

$$s_y^2 = 0.009435$$

$$\text{c.v.} = \frac{s_y}{\bar{y}} 100\% \Rightarrow \text{c.v.} = \frac{0.097133}{6.6} 100\% = 1.47\%$$

| | | $\boxed{\bar{y} = 6.6}$ | | Interval | |
|---|---|---|---|---|---|
| $i$ | $y_i$ | $(y_i - \bar{y})^2$ | Frequency | Lower Bound | Upper Bound |
| 1 | 6.395 | 0.042025 | 1 | 6.36 | 6.40 |
| 2 | 6.435 | 0.027225 | 1 | 6.40 | 6.44 |
| 3 | 6.485 | 0.013225 | | | |
| 4 | 6.495 | 0.011025 | 4 | 6.48 | 6.52 |
| 5 | 6.505 | 0.009025 | | | |
| 6 | 6.515 | 0.007225 | | | |
| 7 | 6.555 | 0.002025 | 2 | 6.52 | 6.56 |
| 8 | 6.555 | 0.002025 | | | |
| 9 | 6.565 | 0.001225 | | | |
| 10 | 6.575 | 0.000625 | 3 | 6.56 | 6.60 |
| 11 | 6.595 | 0.000025 | | | |
| 12 | 6.605 | 0.000025 | | | |
| 13 | 6.615 | 0.000225 | | | |
| 14 | 6.625 | 0.000625 | 5 | 6.60 | 6.64 |
| 15 | 6.625 | 0.000625 | | | |
| 16 | 6.635 | 0.001225 | | | |
| 17 | 6.655 | 0.003025 | | | |
| 18 | 6.655 | 0.003025 | 3 | 6.64 | 6.68 |
| 19 | 6.665 | 0.004225 | | | |
| 20 | 6.685 | 0.007225 | | | |
| 21 | 6.715 | 0.013225 | 3 | 6.68 | 6.72 |
| 22 | 6.715 | 0.013225 | | | |
| 23 | 6.755 | 0.024025 | 1 | 6.72 | 6.76 |
| 24 | 6.775 | 0.030625 | 1 | 6.76 | 6.80 |
| Σ | $\boxed{158.4}$ | $\boxed{0.217000}$ | | | |

# DATA DISTRIBUTION

- The shape with which the given data are spread around the mean
- Histogram provides a simple visual representation of the distribution
  - Constructed by sorting the measurements into intervals

| $i$ | $y_i$ | $(y_i - \bar{y})^2$ | Frequency | Interval Lower Bound | Interval Upper Bound |
|---|---|---|---|---|---|
| 1 | 6.395 | 0.042025 | 1 | 6.36 | 6.40 |
| 2 | 6.435 | 0.027225 | 1 | 6.40 | 6.44 |
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| 23 | 6.755 | 0.024025 | 1 | 6.72 | 6.76 |
| 24 | 6.775 | 0.030625 | 1 | 6.76 | 6.80 |
| $\Sigma$ | 158.4 | 0.217000 | | | |



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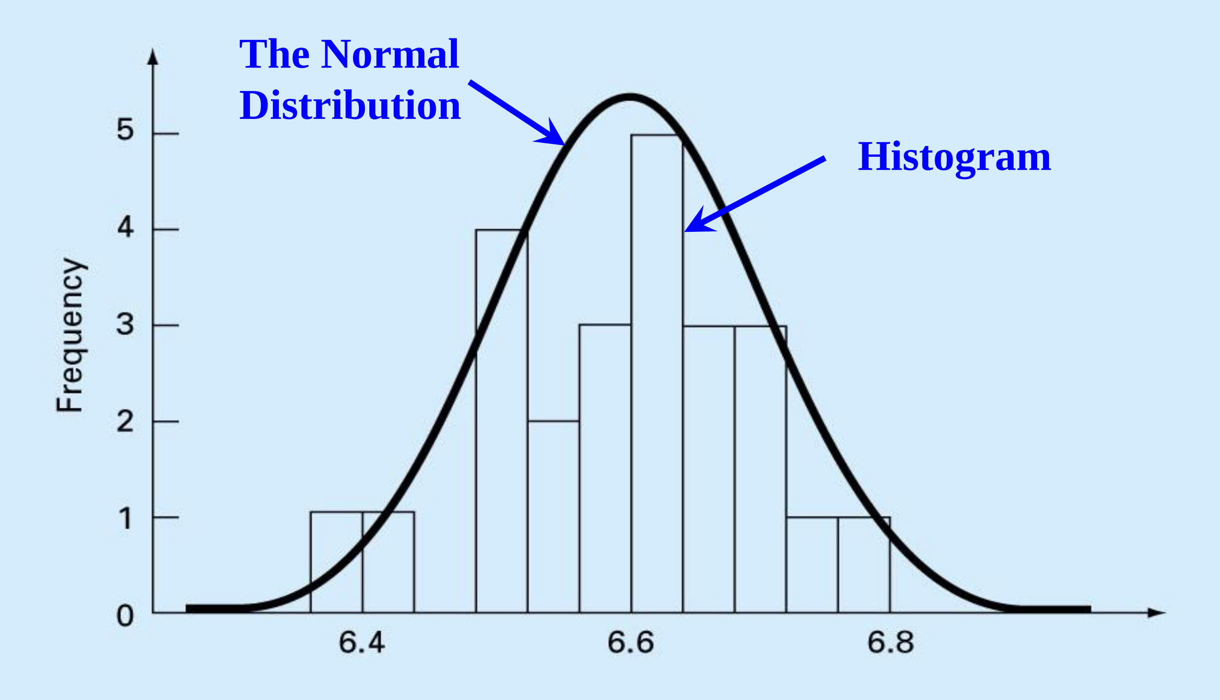
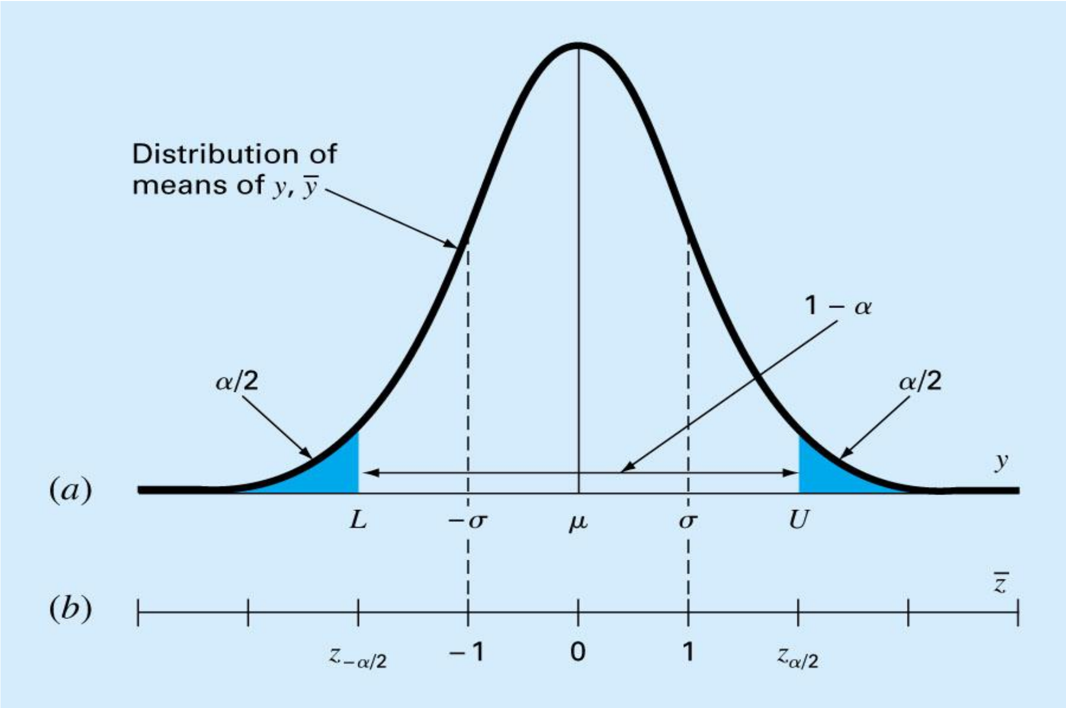
# HISTOGRAM: DISTRIBUTION OF DATA

- Histogram plot
  - X axis: the units of measurement
  - Y axis: frequency of occurrence of each interval
- Histogram often can be approximated by a smooth curve if we have a very large set of data
- The bell-shaped curve is called: Normal distribution



# NORMAL DISTRIBUTION

- If a quantity is normally distributed, the range defined by $\bar{y} - s_y$ to $\bar{y} + s_y$ will encompass approximately 68% of the total measurements
- the range defined by $\bar{y} - 2s_y$ to $\bar{y} + 2s_y$ will encompass approximately 95% of the total measurements
  - $\bar{y} = 6.6, s_y = 0.097133$
  - $\bar{y} - 2s_y$ to $\bar{y} + 2s_y$: 6.405734 to 6.794266
  - If a value is 7.35, it may be erroneous





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